

All-Damping Mehler and Critical-Rate Identities for the Harmonic Kramers Diffusion

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Abstract

We give one exact finite-dimensional theorem for the harmonic Kramers–Langevin diffusion across underdamped, critically damped, overdamped, and zero-damping faces. A single matrix-exponential formula yields the Gaussian Mehler transition and the Lyapunov covariance difference. The Gibbs law, the positive-damping Kalman bracket, stationary correlations, and the drift spectral-abscissa rate are then explicit. Critical damping maximizes the asymptotic exponent but retains a $te^{-\omega t}$ Jordan prefactor. This is a stochastic Markov result, not an arithmetic zeta or a Hilbert–Pólya construction. Independent exact reconstruction (384 assertions), symbolic checks (26 identities), byte replay and 26 hostile mutations accompany the two substantive revisions.

1 Object and boundary atlas

Fix physical time $t \geq 0$ and

$$dQ_t = P_t dt, \quad dP_t = (-\omega^2 Q_t - \gamma P_t) dt + \sqrt{2\gamma/\beta} dW_t, \quad \omega, \beta > 0, \gamma \geq 0. \quad (1)$$

For $X = (Q, P)^T$, let $A = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -\gamma \end{bmatrix}$. The $\gamma = 0$ face is a deterministic Hamiltonian oscillator. The $\omega = 0$ face is kept separate: position is unconfined and there is no finite Gibbs probability with the covariance below.

Theorem 1 (All-damping flow and Gaussian transition). *Put $\alpha = \gamma/2$. If $\alpha < \omega$, set $\nu = (\omega^2 - \alpha^2)^{1/2}$ and $(c, s) = (\cos(\nu t), \sin(\nu t)/\nu)$. If $\alpha = \omega$, set $(c, s) = (1, t)$. If $\alpha > \omega$, set $\delta = (\alpha^2 - \omega^2)^{1/2}$ and $(c, s) = (\cosh(\delta t), \sinh(\delta t)/\delta)$. Then*

$$M_t = e^{tA} = e^{-\alpha t} \begin{pmatrix} c + \alpha s & s \\ -\omega^2 s & c - \alpha s \end{pmatrix}, \quad M_0 = I, \quad \det M_t = e^{-\gamma t}. \quad (2)$$

For $\gamma > 0$ and $t > 0$,

$$X_t \mid X_0 = x \sim \mathcal{N}(M_t x, C_t), \quad C_t = \Sigma - M_t \Sigma M_t^T, \quad \Sigma = \text{diag}\left(\frac{1}{\beta\omega^2}, \frac{1}{\beta}\right), \quad (3)$$

and C_t is positive definite. For $\gamma = 0$, $C_t = 0$ and the law is the Dirac mass at $M_t x$.

Proof. The characteristic polynomial is $\lambda^2 + \gamma\lambda + \omega^2$. Applying the standard exponential formula to $A + \alpha I$, whose square is $(\alpha^2 - \omega^2)I$, gives (2), with the limiting Jordan expression at equality. With $B = (0, \sqrt{2\gamma/\beta})^T$, direct multiplication gives

$$A\Sigma + \Sigma A^T + BB^T = 0. \quad (4)$$

Differentiating $M_t \Sigma M_t^T$ and integrating (4) proves (3). For $\gamma > 0$, the bracket matrix $[B, AB]$ has determinant $-2\gamma/\beta \neq 0$; hence the controllability Gramian, and therefore C_t , is positive definite for every $t > 0$. $\square$

2 Gibbs law, correlations, and damping rate

Equation (4) immediately gives the invariant centered Gaussian density

$$\pi(q, p) = \frac{\beta\omega}{2\pi} \exp\left[-\frac{\beta}{2}(\omega^2 q^2 + p^2)\right]. \quad (5)$$

For $\omega > 0$ and $\gamma > 0$ it is the unique invariant probability. Under this law,

$$\text{Cov}(X_t, X_0) = M_t \Sigma, \quad C_{QQ} = \frac{m_{11}}{\beta\omega^2}, \quad C_{QP} = \frac{m_{12}}{\beta}, \quad C_{PQ} = \frac{m_{21}}{\beta\omega^2}, \quad C_{PP} = \frac{m_{22}}{\beta}. \quad (6)$$

Proposition 1 (Spectral-abscissa rate and critical optimum). *The drift eigenvalues are $-\gamma/2 \pm \sqrt{\gamma^2/4 - \omega^2}$. The spectral-abscissa decay exponent is*

$$r(\omega, \gamma) = \begin{cases} \gamma/2, & 0 \leq \gamma \leq 2\omega, \\ \gamma/2 - \sqrt{\gamma^2/4 - \omega^2}, & \gamma \geq 2\omega. \end{cases} \quad (7)$$

It is maximized at critical damping, where $r = \omega$. At exactly $\gamma = 2\omega$,

$$M_t = e^{-\omega t} [I + t(A + \omega I)], \quad (8)$$

so the polynomial prefactor is genuine: (7) is an asymptotic exponent and not a uniform sharp bound $\|M_t\| \leq Ce^{-\omega t}$. For every $r' < \omega$ such a bound does hold.

Proof. The two roots of the characteristic polynomial give (7). The first branch increases with γ and the second has derivative $1/2 - \gamma/[4\sqrt{\gamma^2/4 - \omega^2}] < 0$ for $\gamma > 2\omega$. The repeated-root formula gives (8), proving the prefactor statement. $\square$

3 Boundary note

The noiseless Hamiltonian face and the unconfined zero-frequency face require separate treatment; no arithmetic determinant is considered in this paper.

References

- [1] H. A. Kramers, “Brownian motion in a field of force,” *Physica* 7 (1940), 284–304.